



# Information Technology Institute

## SOFTWARE REQUIREMENTS SPECIFICATION

Private Teaching Center

Web Application & PWA

Freelancing Job

**Date:** April 25, 2026

**Status:** Pre-Development Review

**Center Scale:** 300–400 Students

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## 1. Project Overview

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This document defines the full software requirements for a private teaching center management platform. The system covers a web application and a mobile application serving four distinct user roles:

- Owner
- Staff (with sub-roles)
- Students
- Parents.

The center base is 300–400 students.

### 1.1 Goals

- Make a Fully Responsive Mobile First App.
- Support language switch: English and Arabic.
- Make everything easy to reach in minimum steps.

## 2. User Roles & Access

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### 2.1 Role Summary

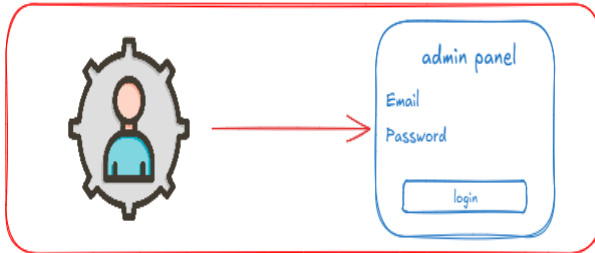
| Role          | Description                                                                   |
|---------------|-------------------------------------------------------------------------------|
| Owner         | Full access to all features.                                                  |
| Staff / Admin | Broad access; manages operations.                                             |
| Teacher       | Access to their own groups, courses, create exams, view student progress.     |
| Student       | Personal dashboard, courses, exams, quizzes, assignments, calendar, payments. |
| Parent        | View child's courses, exams/quiz results, payment summaries.                  |

**Note:** Staff tabs and sidebar navigation items are conditionally shown based on the staff member's assigned role

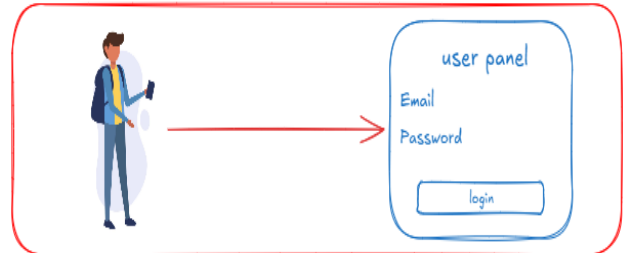
## 2.2 Separate Login Pages

- Staff & Owner: A completely separate, custom login page with Google OAuth and email/password options from the student/parent login.
- Student & Parent: Shared login/register page with Google OAuth and email/password options.

URL/ Login / Admin

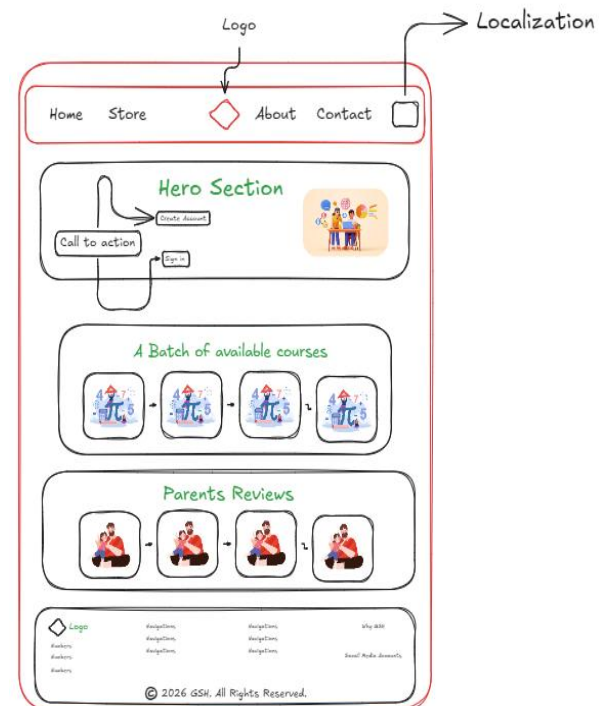


URL/ Login



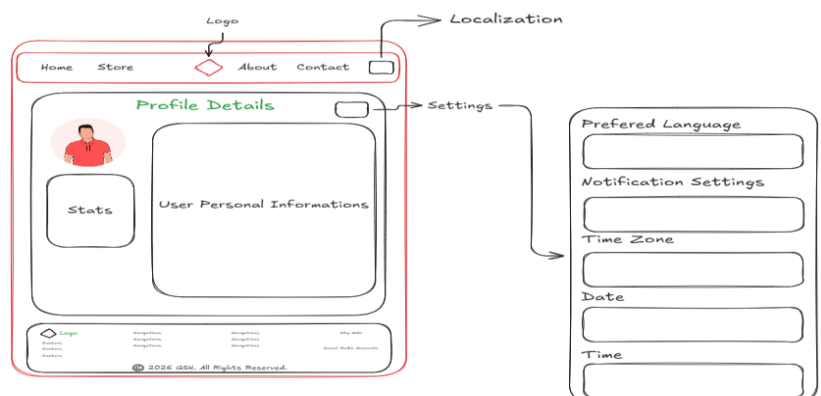
## 3. Landing Page (Public)

- Introduction to the academy and the application.
- Call to action: 'Create Account' button and 'Sign in' button.
- About Academy section.
- Course store preview.
- Language localization toggle: English / Arabic.



## 4. User Page

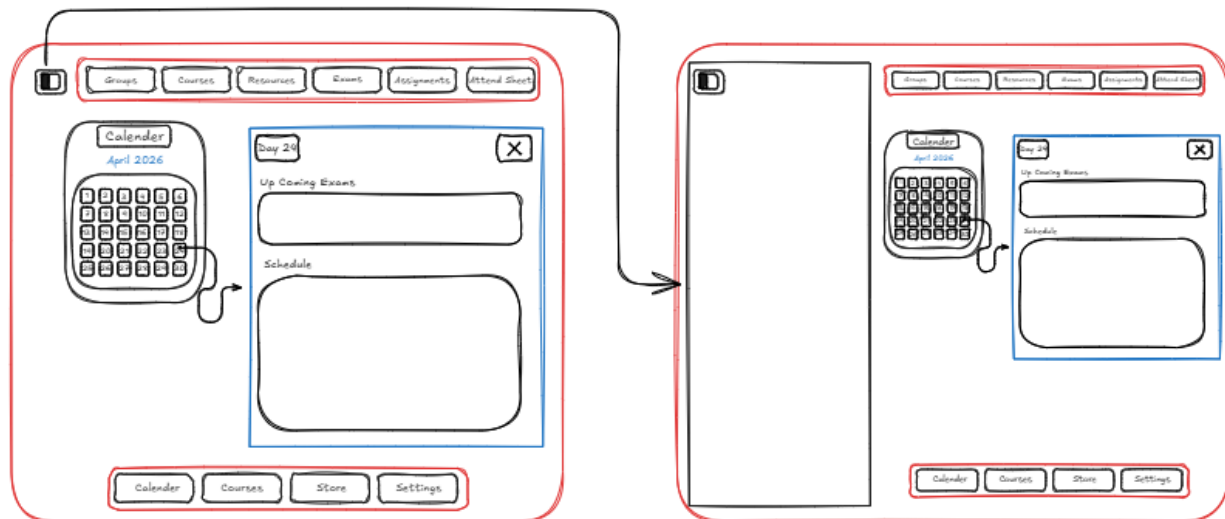
- Profile page for every user type.
- Settings page for every user type.



## 5. Navigation System

### 5.1 Drawer / Sidebar

- left drawer on all dashboards.
- Open/close toggle button.
- Contains quick navigation links to all major sections allowed for the current user role to maximize user experience.



### 5.2 Navigation Bar

- Horizontal scrollable top navigation bar.
- User can navigate using either the drawer or the navbar.



### 5.3 Secondary Navbar / Drawer

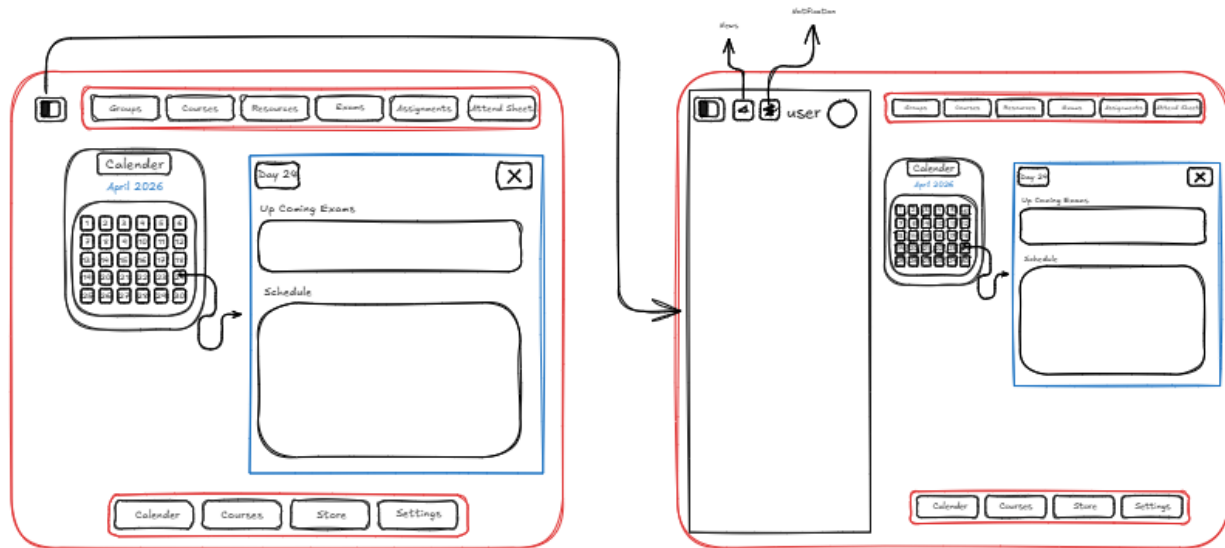
- Nested/sub-pages display a secondary navbar.
- Allows users to navigate back and between nested sections easily.

#### Fast Front/Back Navigation

Home → Course → Lesson

## 5.4 Global Elements on all dashboards

- Notification button: opens a dropdown/popup with recent notifications.
- News button: opens a news/announcement panel.
- Language toggle: English / Arabic.
- Profile quick-access button.
- User general data (Name, Role)



## 6. Owner Dashboard

- The owner has full access to everything. The dashboard provides high-level visibility across management, progress, and revenue.

### 6.1 Overview Widgets

- Total students enrolled.
- Active groups and courses.
- Monthly and cumulative revenue summary.
- Recent payments (cash and Instapay).
- Upcoming exams and quizzes.

## 6.2 System Management (Full Access)

- Student Management
- Group Management
- Course Management
- Payment & Revenue Management
- Exam & Quiz Management
- Staff Management (add, edit, assign roles to staff).
- Recorded Lessons (upload and manage video lessons).
- Logs (full system audit/activity logs).
- Email & WhatsApp notifications to parents
- American Trials Exams
- Reports (progress reports per student, group, or course).

## 7. Staff Dashboard

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- The staff dashboard is identical in structure to the owner dashboard but with tabs and features gated according to the staff member's assigned sub-role (Receptionist, Accountant, Teacher, etc.).

### 7.1 All Staff Can Access

- Their own profile and settings page.
- Notifications and news panel.
- Groups and courses all students assigned to.
- Mark student attendance.
- Mark student as paid.
- Search for available courses and groups.
- View student information.
- Send progress reports via email and WhatsApp to parents.
- View all payment records (cash and Instapay).
- Generate financial reports.

## 8. Student Dashboard

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### 8.1 Home / Overview

- Personal greeting with upcoming schedule.
- Calendar view showing all upcoming exams, quizzes, assignments, and events.
- Notifications and news access.

## 8.2 My Courses

- Current courses with progress indicator.
- Past completed courses.
- Each course shows progress tied to the completed agenda items.
- Attendance sheet per course.
- Course resources (files, links, materials).
- Recorded lesson videos for enrolled courses.

## 8.3 Course Subscription / Store

- Browse all available courses (online or offline).
- Each course listing includes:
  - Video introduction from the teacher.
  - Full agenda of topics to be covered.
  - List of quizzes and exams associated with the course.
  - Course price.
  - Online vs. offline mode.

**Note:** Subscribe to a course. Payment required to confirm enrollment.

## 8.4 Exams & Quizzes

- List of all upcoming exams and quizzes.
- Take an exam or quiz online (automatic assessment after submission).
- History of all past exams and quizzes with scores and progress breakdown.

## 8.5 Assignments

- View assigned tasks and submission status.

## 8.6 My Groups

- All groups the student belongs to.

## 8.7 Calendar

- calendar showing all upcoming events: exams, quizzes, assignments, sessions.
- Filtered to only show items relevant to the student's enrolled courses and groups.

## 9. Parent Dashboard

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- View all children linked to the parent account.
- Child data view:
  - Courses the child is enrolled in (current and past).
  - Exam and quiz results and scores.
  - Assessment breakdown.
  - Attendance records.
  - Payment details: cost per course and monthly summary.
- Notification and news panel.

## 10. Staff Student Management

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- Create, edit, and deactivate student profiles.
- Link parent accounts to student profiles.
- Assign students to groups.
- View full student information and history.
- Search and filter students.
- View attendance records per student.
- View payment history per student.
- Mark attendance (manual entry by staff).

## 11. Staff Group Management

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- Create, edit, and delete groups.
- Assign courses and teachers to groups.
- Assign students to groups.
- Set group schedules.
- View list of all groups and their enrolled students.
- Restrict exam/quiz access to specific groups.

## 12. Staff Course Management

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- Create, edit, and deactivate courses.
- Set course type: Online or Offline.
- Upload teacher introduction video per course.
- Define curriculum/agenda (list of topics; student progress tracked against this).
- Set course price.
- Attach resources (files, documents, links).
- Upload recorded lesson videos.
- Associate quizzes and exams with a course.
- Mark agenda/curriculum items as completed (updates student progress automatically).



## 13. Staff Payment Management

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### 13.1 Payment Methods

- Cash.
- Instapay by scanning the center's Instapay QR code, or sends to the center's Instapay email/number.

### 13.2 Payment Confirmation Flow

1. Student or parent makes payment (cash or Instapay).
2. Student or parent uploads a screenshot of the payment receipt.
3. Staff reviews the screenshot and marks the payment as confirmed.
4. Student's enrollment or exam registration is activated.

### 13.3 Payment Records

- View all payments (filterable by date, student, course, method).
- Revenue reports (daily, monthly, yearly).
- Outstanding balances tracking.

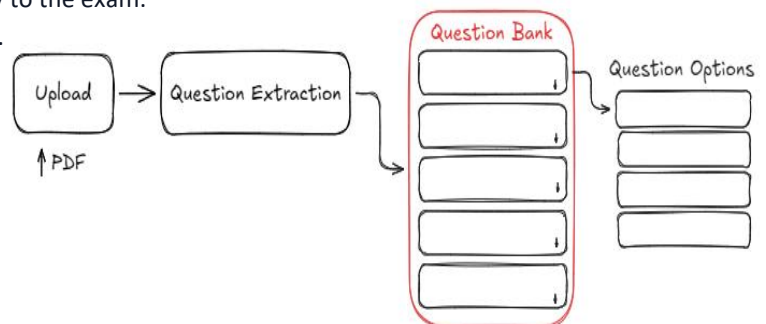
## 14. Staff Exam & Quiz Management

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### 14.1 Exam Creation Flow

Staff or teacher opens the Exam Creation page and completes the following steps:

1. Fill in exam info such as: title, description, duration, assigned course/group, date.
2. Upload exam PDF.
3. Extracted questions and options.
4. Add extracted questions appeared as expandable cards in a Question Bank.
5. Staff builds the exam by:
  - a. Dragging and dropping a question card from the bank into the exam.
  - b. Then dragging and dropping the answer options one by one into the question.
  - c. They can type the question and options manually.
  - d. They can go in hybrid (some dragged, some typed).
6. After all questions are added, assign an access key to the exam.
7. Select which groups are allowed to take the exam.
8. Schedule the exam.
9. Save the exam.



## 14.2 Taking an Exam (Student)

- Student enters the exam password to enroll to it.
- Student answers all questions within the set duration.
- Exam ends and auto submits after the exam time.
- Exam is automatically graded after submission.
- Score are shown instantly.

## 14.3 Exam & Quiz Records

- Full history of all exams and quizzes per student.
- Score breakdown per question.
- Progress analysis over time.
- Visible in both student and parent dashboards.

## 15. American Trials Exams — Special (Search for it)

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A custom store page within the platform for list of 12 American Trials standardized exams.

- Each exam is listed with its specific price.
- Student selects an exam and fills in a registration form.
- Student pays the exam price.
- after payment confirmation, the student is registered and can take the exam at specific time the center decides and send an email and message with the exam data.

## 16. Communication & Notifications

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### 16.1 Dashboard Notifications

- Notification bell icon on all dashboards.
- popup showing recent notifications.

### 16.2 News / Announcements

- Separate 'News' button on all dashboards.
- Admin can post news and announcements visible to all or specific roles.

### 16.3 Email to Parents

- Staff can send a structured, templated email to parents.
- Content includes: child's progress, assessment results, if child absent, and any updates during the course.
- Dedicated 'Send Progress Report' page with pre-filled student data.

## 16.4 WhatsApp to Parents

- Staff can send WhatsApp messages to parents directly from the platform.
- Used for quick updates, payment reminders, schedule changes.

## 17. Additional Pages

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### 17.1 Store

- Public-facing course catalog.
- Course detail page with teacher video, curriculum, price, and enrollment CTA.
- American Trials Exams listed with pricing and registration.

### 17.2 About Academy

- Public page describing the center, its mission, teachers, and why students should choose it.

### 17.3 Settings

- Available to all user roles.
- Includes: language preference, Notification, Time Zone, Date, Time.

### 17.4 Profile

- Every user has a profile page.
- Personal information editable (critical information disabled), profile photo, contact details.

### 17.5 Logs (Search for it)

- Owner and staff can view logs.
- Logs include: logins, payments, exam submissions, record edits.

## 18. Progressive Web App (PWA)

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- Make it installable on mobile and desktop using PWA.
- An 'Install App' button is shown to users.
- The button automatically disappears once the app has been installed.
- Supports offline-capable features where applicable.

## 19. Mobile Application

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### 19.1 Must Have

- Screenshot prevention.
- Published to the Google Play Store.
- Download link for the APK is available directly from the website.

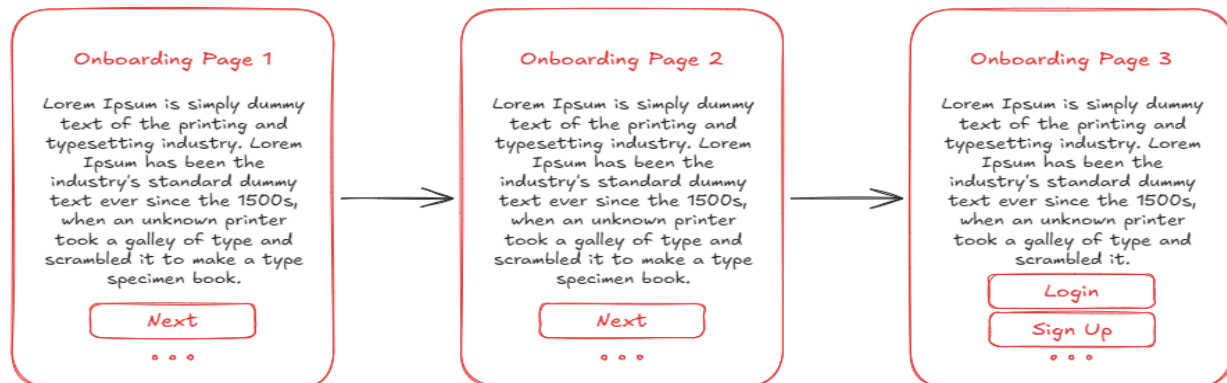
### 19.2 Splash Screen

- Shown for 2–3 seconds when the app is launched and Branded with the academy's Logo.

### 19.3 Onboarding Flow (First Launch Only)

- The onboarding is displayed only the first time the user opens the app, or if they are not logged in. It consists of 3 screens:
1. **Page 1:** Who We Are with an introduction to the academy.
  2. **Page 2:** Why Choose Us with key benefits and differentiators.
  3. **Page 3:** Get Started with a call to action with two buttons: Log in and Register.

**Note:** will not be shown again on subsequent launches.



## 20. Design & UX Requirements

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- Modern, clean, Professional design.
- Mobile-first, fully responsive design for all screen sizes.
- Everything easy to reach.
- English and Arabic.